Design Explanation & Layout Report

The goal of this project was to design a reliable, clean, and custom printed circuit board (PCB) built to work alongside an STM32 microcontroller development board, of which I used the “black pill” as my primary design reference. The power system includes a 3.3V voltage regulator to step down an external 5V supply into a stable operating rail. The overall design also integrates various components including push buttons, a potentiometer, LEDs, a buzzer, and expansion headers for I2C and UART. Every schematic symbol was carefully matched with the correct physical footprint size to ensure a smooth and accurate hardware build.

The schematic design process began with a layout strategy, keeping the power regulation circuitry cleanly separated on one side. For the remaining components, I mapped and connected them to pin sockets designed to match the specific pinout configuration of my STM32 Black Pill reference. To ensure accuracy, I researched the exact functions of each pin on the development board—covering digital GPIOs, analog ADC channels, communication interfaces like I2C and UART, and core power pins before wiring my components to their corresponding socket locations. During this phase, I encountered power error messages (power pin not driven). Even after restarting the design from scratch and introducing a power flag at the 5V input, the error persisted. Through research, I discovered that I needed to place a second PWR_FLAG specifically on the ground (GND) net in addition to the power net, which finally allowed the design software to recognize the power distribution properly.

Once the schematic was finalized, I moved on to component placement and layout planning. I updated the PCB layout directly from the schematic and arranged the components carefully, rotating and moving them to untangle the connection ratsnest as much as possible for a less tangled layout. The physical board dimensions were then defined by drawing the outline on the Edge.Cuts layer. For surface-mount devices like capacitors and LEDs, no holes are required on the board since they solder directly onto surface pads, whereas connectors use standard drilled holes.

To optimize routing and minimize complexity, I used the entire bottom layer of the board as a ground plane. This meant I only needed a single via to connect straight down to the ground plane, removing the need for complex routing. Furthermore, I carefully adjusted my track widths, making the power tracks twice as wide as the standard signal tracks to safely handle higher current loads. To ensure voltage stability and remove electrical noise, decoupling capacitors were placed as close as possible to the regulator, providing essential filtering and bulk power storage for a clean output.

Finally, the design underwent verification through the Electrical Rules Check (ERC) and the Design Rules Check (DRC), both of which successfully completed with zero errors and zero warnings. All manufacturing file, thus, Gerber layers, and the Bill of Materials (BOM) are attached